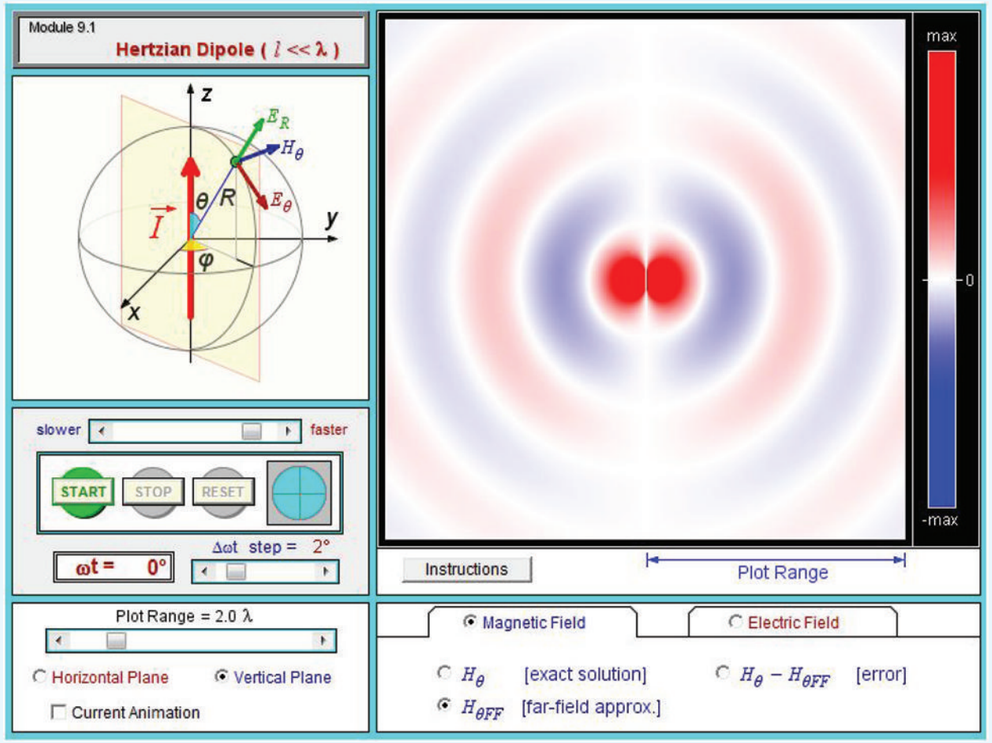
Module 9.1 Hertzian Dipole ($l \ll \lambda$) For a short dipole oriented along the z axis, this module displays the field distributions for **E** and **H** in both the horizontal and vertical planes. It can also animate the radiation process and current flow through the dipole.



where S is the radial component of the time-average Poynting vector $\mathbf{S}_{av}$. In the far-field region of any antenna, $\mathbf{S}_{av}$ is always in the radial direction. In a spherical coordinate system,

$$dA = R^2 \sin\theta \, d\theta \, d\phi, \tag{9.17}$$

and the ***solid angle*** $d\Omega$ associated with dA, which is defined as the subtended area divided by R^2, is given by

$$d\Omega = \frac{dA}{R^2} = \sin\theta \, d\theta \, d\phi \quad \text{(sr)}. \tag{9.18}$$

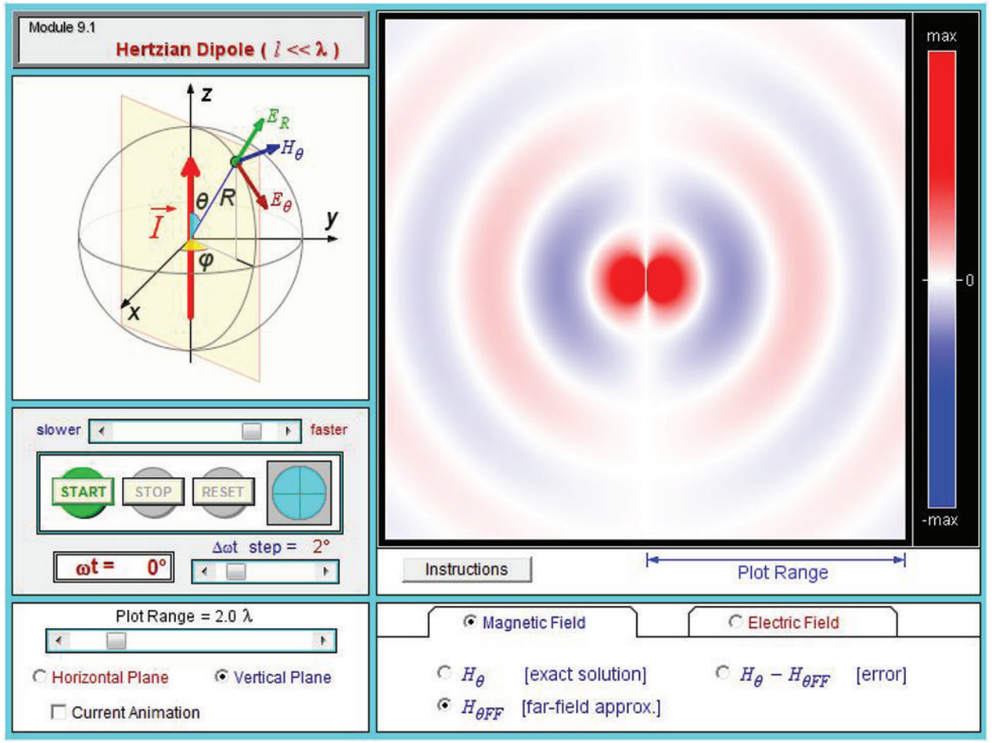
Note that, whereas a planar angle is measured in radians and the angular measure of a complete circle is 2π (rad), a solid angle is measured in ***steradians*** (sr), and the angular measure for a spherical surface is $\Omega = (4\pi R^2)/R^2 = 4\pi$ (sr). The solid angle of a hemisphere is 2π (sr).

Using the relation $dA = R^2 \, d\Omega$, dP_{rad} can be rewritten as

$$dP_{\text{rad}} = R^2 S(R, \theta, \phi) \, d\Omega. \tag{9.19}$$

The total power radiated by an antenna through a spherical surface at a fixed distance R is obtained by integrating

模块 9.1 赫兹偶极子 (l ≪ λ) 对于沿 z 轴定向的短偶极子，此模块显示 E 和 H 在水平和垂直平面上的场分布。它还可以动画化辐射过程和流经偶极子的电流。



其中 S 是时间平均坡印廷向量 S_{av} 的径向分量。在任何天线的远场区域，S_{av} 始终处于径向方向。在球坐标系中，

$$dA = R^2 \sin\theta \, d\theta \, d\phi, \tag{9.17}$$

与 dA 相关的立体角 dΩ 定义为所对面积除以 R^2，由下式给出

$$d_{\Omega} = dAR_2 = \text{罪恶}\theta \; d\theta \; d\phi \quad \text{(sr)。} \tag{9.18}$$

请注意，平面角以弧度为单位测量，完整圆的角度测量为 2π (rad)，而立体角以球面度 (sr) 测量，球面的角度测量为 $\Omega = (4\pi R^2)/R^2 = 4\pi$ (sr)。半球的立体角为2π (sr)。

使用关系 $dA = R^2 \, d\Omega$，dP_{rad} 可以重写为

$$dP_{\text{rad}} = R^2 S(R, \theta, \phi) \, d\Omega. \tag{9.19}$$

天线在固定距离 R 处通过球面辐射的总功率可通过积分获得

Eq. (9.19) over that surface:

$$P_{\text{rad}} = R^2 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} S(R,\theta,\phi) \sin\theta \, d\theta \, d\phi$$
$$= R^2 S_{\max} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} F(\theta,\phi) \sin\theta \, d\theta \, d\phi$$
$$= R^2 S_{\max} \iint_{4\pi} F(\theta,\phi) \, d\Omega \qquad \text{(W)}, \tag{9.20}$$

where $F(\theta,\phi)$ is the normalized radiation intensity defined by Eq. (9.13). The 4π symbol under the integral sign is used as an abbreviation for the indicated limits on θ and ϕ. Formally, P_{rad} is called the ***total radiated power***.

9-2.1 Antenna Pattern

Each specific combination of the zenith angle θ and the azimuth angle ϕ denotes a specific direction in the spherical coordinate system of **Fig. 9-8**. The normalized radiation intensity $F(\theta,\phi)$ characterizes the directional pattern of the energy radiated by an antenna, and a plot of $F(\theta,\phi)$ as a function of both θ and ϕ constitutes a three-dimensional pattern—an example of which is shown in **Fig. 9-9**.

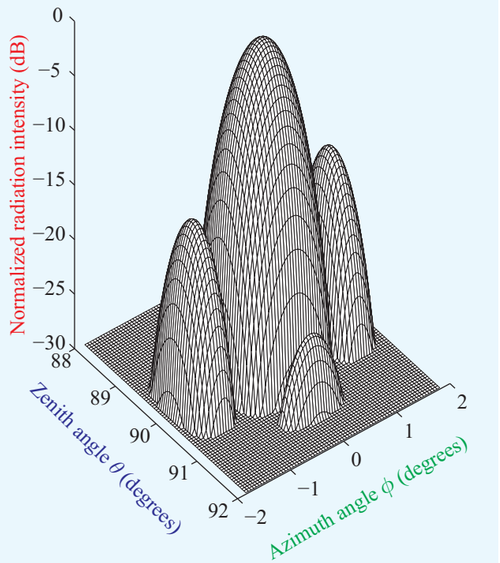


Figure 9-9 Three-dimensional pattern of a narrow-beam antenna.

Often, it is of interest to characterize the variation of $F(\theta,\phi)$ in the form of two-dimensional plots in specific planes in the spherical coordinate system. The two planes most commonly specified for this purpose are the elevation and azimuth planes. The ***elevation plane***, also called the θ plane, is a plane corresponding to a constant value of ϕ. For example, $\phi = 0$ defines the x–z plane and $\phi = 90°$ defines the y–z plane, both of which are elevation planes (**Fig. 9-8**). A plot of $F(\theta,\phi)$ versus θ in either of these planes constitutes a two-dimensional pattern in the elevation plane. This is not to imply, however, that the elevation-plane pattern is necessarily the same in all elevation planes.

The ***azimuth plane***, also called the ϕ plane, is specified by $\theta = 90°$ and corresponds to the x–y plane. The elevation and azimuth planes are often called the two ***principal planes*** of the spherical coordinate system.

Some antennas exhibit highly directive patterns with narrow beams, in which case it is often convenient to plot the antenna pattern on a decibel scale by expressing F in decibels:

$$F \text{ (dB)} = 10 \log F.$$

As an example, the antenna pattern shown in **Fig. 9-10(a)** is plotted on a decibel scale in polar coordinates with its intensity as the radial variable. This format permits a convenient visual interpretation of the directional distribution of the ***radiation lobes***.

Another format commonly used for inspecting the pattern of a narrow-beam antenna is the rectangular display shown in **Fig. 9-10(b)**, which permits the pattern to be easily expanded by changing the scale of the horizontal axis. These plots represent the variation in only one plane in the observation sphere: the $\phi = 0$ plane. Unless the pattern is symmetrical in ϕ, additional patterns are required to define the overall variation of $F(\theta,\phi)$ with θ and ϕ.

Strictly speaking, the polar angle θ is always positive, being defined over the range from 0° (z direction) to 180° ($-z$ direction), yet the θ axis in **Fig. 9-10(b)** is shown to have both positive and negative values. This is not a contradiction, but rather a different form of plotting antenna patterns. The right-hand half of the plot represents the variation of F (dB) with θ because θ is increased in a clockwise direction in the x–z plane [see inset in **Fig. 9-10(b)**], corresponding to $\phi = 0$. The left-hand half of the plot represents the variation of F (dB) with θ as θ is increased in a counterclockwise direction at $\phi = 180°$. Thus, a negative θ value simply denotes that the direction (θ,ϕ) is in the left-hand half of the x–z plane.

The pattern shown in **Fig. 9-10(a)** indicates that the antenna is fairly directive, since most of the energy is radiated through

等式。 (9.19) 在该表面上：

$$P_{\text{rad}} = R^2 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} S(R,\theta,\phi) \sin\theta \, d\theta \, d\phi$$
$$= R^2 S_{\max} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} F(\theta,\phi) \sin\theta \, d\theta \, d\phi$$
$$= R^2 S_{\max} \iint_{4\pi} F(\theta,\phi) \, d\Omega \qquad \text{(W)}, \tag{9.20}$$

其中 F(θ , ϕ) 是由方程式定义的归一化辐射强度。 (9.13) 。积分符号下的 4π 符号用作 θ 和 ϕ 指示极限的缩写。形式上，P_{rad} 称为总辐射功率。

9-2.1 天线方向图

天顶角θ和方位角ϕ的每个特定组合表示图9-8的球面坐标系中的特定方向。归一化辐射强度 F(θ , ϕ) 表征了天线辐射能量的方向图，F(θ , ϕ) 作为 θ 和 ϕ 的函数的绘图构成了一个三维图，其示例如图 9-9 所示。

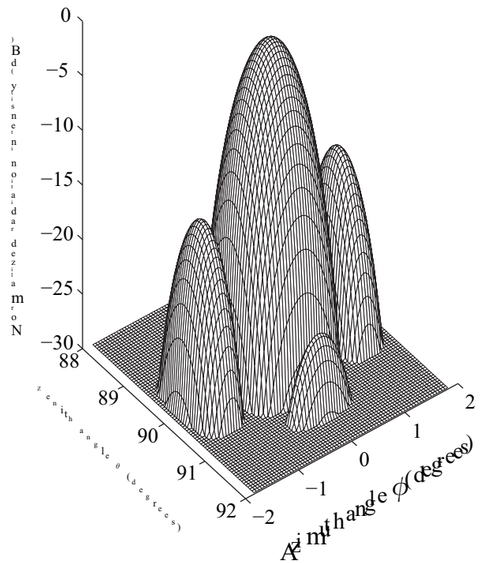


图9-9 窄波束天线的三维方向图。

通常，人们感兴趣的是在球坐标系中的特定平面上以二维图的形式描述 F(θ , ϕ) 的变化。为此目的最常指定的两个平面是仰角平面和方位角平面。高程平面也称为θ平面，是与常数值ϕ相对应的平面。例如，ϕ = 0 定义 x-z 平面，ϕ = 90° 定义 y-z 平面，这两个平面都是高程平面（图 9-8）。这些平面中任一平面中的 F(θ , ϕ) 与 θ 的关系图构成了高程平面中的二维图案。然而，这并不意味着所有高程平面中的高程平面图案必须相同。

方位角平面，也称为 ϕ 平面，由 θ = 90° 指定，对应于 x–y 平面。仰角平面和方位角平面通常被称为球坐标系的两个主平面。

一些天线表现出具有窄波束的高度定向方向图，在这种情况下，通过以分贝表示 F，通常可以方便地在分贝刻度上绘制天线方向图：

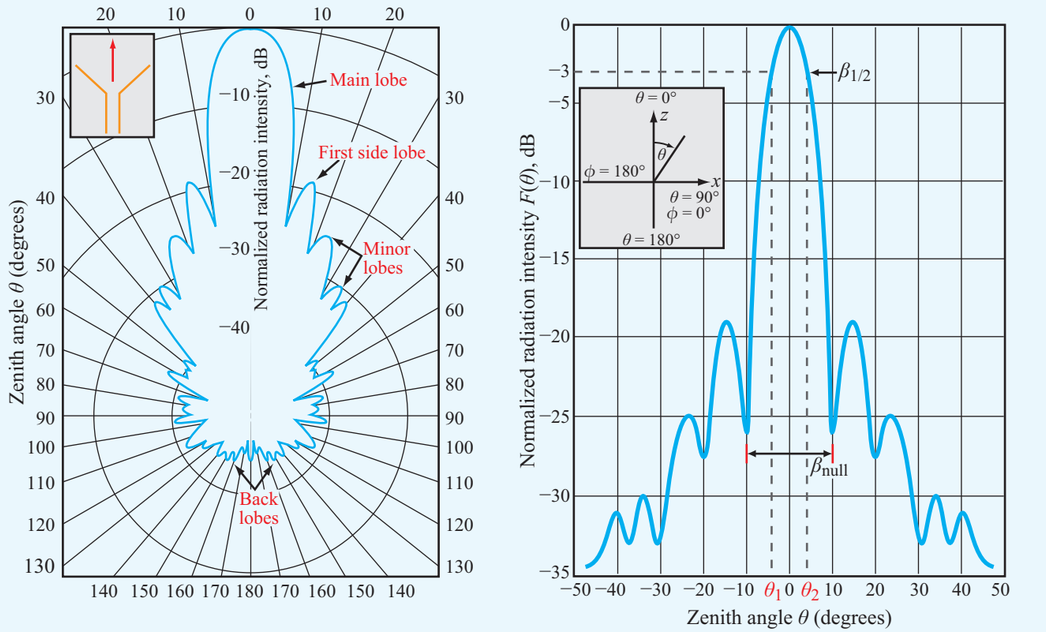
$$F \text{ (dB)} = 10 \log F.$$

作为一个例子，图9-10(a)所示的天线方向图是在极坐标中以分贝刻度绘制的，其强度作为径向变量。这种格式可以方便地直观地解释辐射瓣的方向分布。

另一种常用的检查窄波束天线方向图的格式是图 9-10(b) 所示的矩形显示，它允许通过改变水平轴的比例来轻松扩展方向图。这些图仅表示观测范围内一个平面的变化：ϕ = 0 平面。除非模式在 ϕ 中是对称的，否则需要额外的模式来定义 F(θ , ϕ) 与 θ 和 ϕ 的整体变化。

严格来说，极角θ始终为正，定义范围为0° (z方向)到180° (−z方向)，但图9-10(b)中的θ轴显示为既有正值又有负值。这并不矛盾，而是绘制天线方向图的不同形式。图的右半部分表示 F (dB) 随 θ 的变化，因为 θ 在 x-z 平面中沿顺时针方向增加[参见图 9-10(b) 中的插图]，对应于 ϕ = 0。图的左半部分表示 F (dB) 随 θ 的变化，因为 θ 在 ϕ = 处沿逆时针方向增加180°。因此，负 θ 值仅表示方向 (θ , ϕ) 位于 x–z 平面的左半部分。

图 9-10(a) 所示的图案表明天线具有相当的方向性，因为大部分能量是通过



(a) Polar diagram (b) Rectangular plot

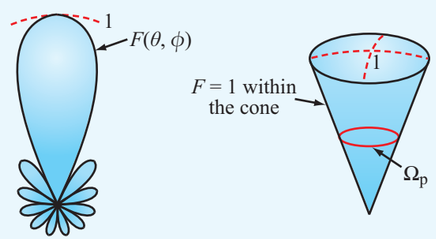
Figure 9-10 Representative plots of the normalized radiation pattern of a microwave antenna in (a) polar form and (b) rectangular form.

a narrow sector called the ***main lobe***. In addition to the main lobe, the pattern exhibits several ***side lobes*** and ***back lobes*** as well. For most applications, these extra lobes are considered undesirable because they represent wasted energy for transmitting antennas and potential interference directions for receiving antennas.

9-2.2 Beam Dimensions

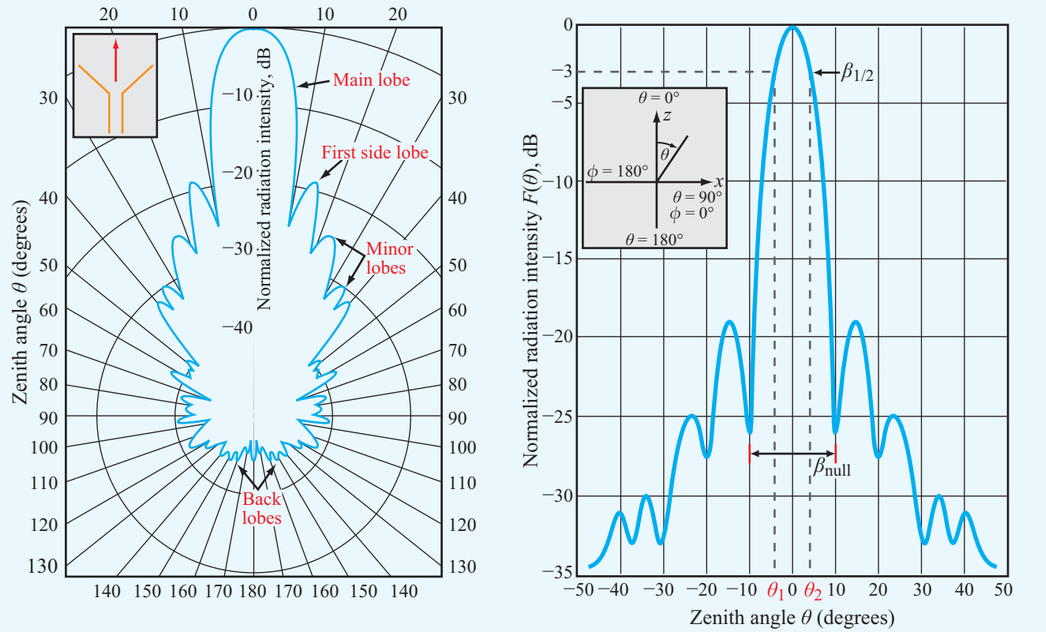
For an antenna with a single main lobe, the ***pattern solid angle*** Ω_p describes the equivalent width of the main lobe of the antenna pattern (**Fig. 9-11**). It is defined as the integral of the normalized radiation intensity $F(\theta, \phi)$ over a sphere:

$$\Omega_p = \iint_{4\pi} F(\theta, \phi)\, d\Omega \quad \text{(sr)}. \tag{9.21}$$



(a) Actual pattern (b) Equivalent solid angle

Figure 9-11 The pattern solid angle Ω_p defines an equivalent cone over which all of the radiation of the actual antenna is concentrated with uniform intensity equal to the maximum of the actual pattern.



(a) Polar diagram (b) Rectangular plot

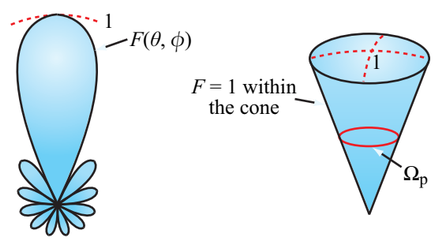
图9-10 微波天线归一化辐射方向图的两种表示：(a) 极坐标图；(b) 直角坐标图。

称为主瓣的狭窄部分。除了主瓣之外，该模式还表现出多个旁瓣和后瓣。对于大多数应用来说，这些额外的波瓣被认为是不受欢迎的，因为它们代表了发射天线的能量浪费和接收天线的潜在干扰方向。

9-2.2 波束立体角

对于单主瓣天线，波束立体角 Ω_p描述了天线方向图主瓣的等效宽度（图9-11）。它被定义为球体上归一化辐射强度 F(θ , ϕ) 的积分：

$$\Omega_p = \iint_{4\pi} F(\theta, \phi)\, d\Omega \quad \text{(sr)}. \tag{9.21}$$



(a) Actual pattern (b) Equivalent solid angle

图 9-11 波束立体角 Ω_p 定义了一个等效圆锥体，实际天线的所有辐射都集中在该圆锥体上，其均匀强度等于实际方向图的最大值。

► For an isotropic antenna with $F(\theta,\phi)=1$ in all directions, $\Omega_p = 4\pi$ (sr). ◄

The pattern solid angle characterizes the directional properties of the three-dimensional radiation pattern. To characterize the width of the main lobe in a given plane, the term used is ***beamwidth***. The ***half-power beamwidth***, or simply the beamwidth β, is defined as the angular width of the main lobe between the two angles at which the magnitude of $F(\theta,\phi)$ is equal to half of its peak value (or -3 dB on a decibel scale). For example, for the pattern displayed in **Fig. 9-10(b)**, β is given by

$$\beta = \theta_2 - \theta_1, \tag{9.22}$$

where θ_1 and θ_2 are the ***half-power angles*** at which $F(\theta,0)=0.5$ (with θ_2 denoting the larger value and θ_1 denoting the smaller one, as shown in the figure). If the pattern is symmetrical and the peak value of $F(\theta,\phi)$ is at $\theta=0$, then $\beta = 2\theta_2$. For the short-dipole pattern shown earlier in **Fig. 9-7(a)**, $F(\theta)$ is maximum at $\theta = 90°$, θ_2 is at 135°, and θ_1 is at 45°. Hence, $\beta = 135° - 45° = 90°$. The beamwidth β is also known as the ***3 dB beamwidth***. In addition to the half-power beamwidth, other beam dimensions may be of interest for certain applications, such as the ***null beamwidth*** β_{null}, which is the angular width between the first nulls on the two sides of the peak (**Fig. 9-10(b)**).

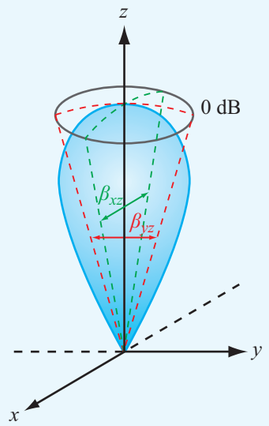


Figure 9-12 The solid angle of a unidirectional radiation pattern is approximately equal to the product of the half-power beamwidths in the two principal planes; that is, $\Omega_p \approx \beta_{xz}\beta_{yz}$.

9-2.3 Antenna Directivity

The ***directivity*** D of an antenna is defined as the ratio of its maximum normalized radiation intensity, $F_{\max}$ (which by definition is equal to 1), to the average value of $F(\theta,\phi)$ over all directions (4π space):

$$D = \frac{F_{\max}}{F_{\text{av}}} = \frac{1}{\frac{1}{4\pi}\iint_{4\pi} F(\theta,\phi)\,d\Omega} = \frac{4\pi}{\Omega_p}. \tag{9.23}$$

(dimensionless)

Here Ω_p is the pattern solid angle defined by Eq. (9.21). Thus, the narrower Ω_p of an antenna pattern is, the greater is the directivity. For an isotropic antenna, $\Omega_p = 4\pi$; hence, its directivity $D_{\text{iso}} = 1$.

By using Eq. (9.20) in Eq. (9.23), D can be expressed as

$$D = \frac{4\pi R^2 S_{\max}}{P_{\text{rad}}} = \frac{S_{\max}}{S_{\text{av}}}, \tag{9.24}$$

where $S_{\text{av}} = P_{\text{rad}}/(4\pi R^2)$ is the average value of the radiated power density and is equal to the total power radiated by the antenna, P_{rad}, divided by the surface area of a sphere of radius R.

► Since $S_{\text{av}} = S_{\text{iso}}$, where S_{iso} is the power density radiated by an isotropic antenna, D represents the ratio of the maximum power density radiated by the antenna to the power density radiated by an isotropic antenna, where both are measured at the same range R and excited by the same amount of input power. ◄

Usually, D is expressed in decibels:* D (dB) $= 10\log D$.

For an antenna with a single main lobe pointing in the z direction shown in **Fig. 9-12**, Ω_p may be approximated as the product of the half-power beamwidths β_{xz} and β_{yz} (in radians):

$$\Omega_p \approx \beta_{xz}\beta_{yz}, \tag{9.25}$$

and

$$D = \frac{4\pi}{\Omega_p} \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}. \quad \textbf{(single main lobe)} \tag{9.26}$$

*A note of caution: Even though we often express certain dimensionless quantities in decibels, we should always convert their decibel values to natural values before using them in the relations given in this chapter.

► 对于各方向 $F(\theta,\phi)=1$ 的各向同性天线，$\Omega_p = 4\pi$ (sr)。 ◄

方向图立体角表征了三维辐射方向图的方向特性。为了表征给定平面中主瓣的宽度，使用的术语是波束宽度。半功率波束宽度，或简称波束宽度 β，定义为两个角度之间主瓣的角宽度，在该角度处 $F(\theta,\phi)$ 的幅度等于其峰值的一半（或分贝标度上的 −3 dB）。例如，对于图 9-10(b) 中显示的模式，β 由下式给出

$$\beta = \theta_2 - \theta_1, \tag{9.22}$$

其中，θ_1和θ_2是$F(\theta,0)=0.5$时的半功率角（其中θ_2表示较大值，θ_1表示较小值，如图所示）。如果图案是对称的并且 $F(\theta,\phi)$ 的峰值位于 $\theta=0$，则 $\beta = 2\theta_2$。对于图 9-7(a) 中所示的短偶极子模式，$F(\theta)$ 在 $\theta = 90°$ 处最大，θ_2 在 135° 处最大，θ_1 在 45° 处最大。因此，$\beta = 135° - 45° = 90°$。波束宽度 β 也称为 3 dB 波束宽度。除了半功率波束宽度之外，某些应用还可能关注其他波束尺寸，例如零波束宽度 β_{null}，它是峰两侧第一个零点之间的角宽度（图 9-10(b)）。

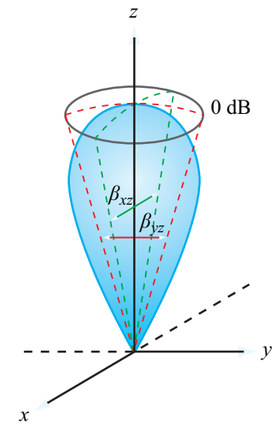


图9-12 单向辐射方向图的立体角约等于两个主平面内半功率波束宽度的乘积；即$\Omega_p \approx \beta_{xz}\beta_{yz}$。

9-2.3 天线方向性

天线的方向性 D 定义为其最大归一化辐射强度 $F_{\max}$（（根据定义等于 1））与所有方向（4π 空间）上 $F(\theta,\phi)$ 平均值的比率：

$$D = \frac{F_{\max}}{F_{\text{av}}} = \frac{1}{\frac{1}{4\pi}\iint_{4\pi} F(\theta,\phi)\,d\Omega} = \frac{4\pi}{\Omega_p}. \tag{9.23}$$

(dimensionless)

这里Ω_p是由方程定义的图案立体角。（9.21）。因此，天线方向图Ω_p越窄，方向性越大。对于各向同性天线，$\Omega_p = 4\pi$；因此，其方向性 $D_{\text{iso}} = 1$。

通过使用等式。(9.20) 式(9.23), D 可表示为

$$D = \frac{4\pi R^2 S_{\max}}{P_{\text{rad}}} = \frac{S_{\max}}{S_{\text{av}}}, \tag{9.24}$$

其中 $S_{\text{av}} = P_{\text{rad}}/(4\pi R^2)$ 是辐射功率密度的平均值，等于辐射的总功率

► 由于 $S_{\text{av}} = S_{\text{iso}}$（其中 S_{iso} 是各向同性天线辐射的功率密度），D 表示天线辐射的最大功率密度与各向同性天线辐射的功率密度之比，其中两者都是在相同的距离 R 处测量并由相同的输入功率激励。 ◄

通常，D以分贝表示：* D (dB) = 10 log D。

对于图 9-12 所示的具有指向 z 方向的单主瓣的天线，Ω_p 可以近似为半功率波束宽度 β_{xz} 和 β_{yz}（（以弧度表示））的乘积：

$$\Omega_p \approx \beta_{xz}\beta_{yz}, \tag{9.25}$$

和

$$D = \frac{4\pi}{\Omega_p} \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}. \quad \textbf{(single main lobe)} \tag{9.26}$$

*A note of caution: Even though we often express certain dimensionless quantities in decibels, we should always convert their decibel values to natural values before using them in the relations given in this chapter.

Although approximate, this relation provides a useful method for estimating the antenna directivity from measurements of the beamwidths in the two orthogonal planes whose intersection is the axis of the main lobe.

Example 9-1: Antenna Radiation Properties

Determine (a) the direction of maximum radiation, (b) pattern solid angle, (c) directivity, and (d) half-power beamwidth in the y–z plane for an antenna that radiates only into the upper hemisphere with normalized radiation intensity given by $F(\theta,\phi) = \cos^2\theta$.

Solution: The statement that the antenna radiates through only the upper hemisphere is equivalent to

$$F(\theta,\phi) = F(\theta) = \begin{cases} \cos^2\theta & \text{for } 0 \le \theta \le \pi/2 \\ & \text{and } 0 \le \phi \le 2\pi, \\ 0 & \text{elsewhere.} \end{cases}$$

(a) The function $F(\theta) = \cos^2\theta$ is independent of ϕ and is maximum when $\theta = 0°$. A polar plot of $F(\theta)$ is shown in **Fig. 9-13**.

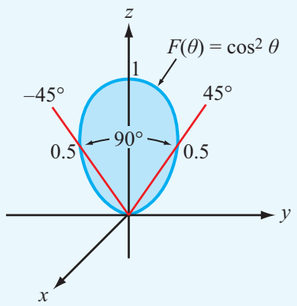


Figure 9-13 Polar plot of $F(\theta) = \cos^2\theta$.

(b) From Eq. (9.21), the pattern solid angle Ω_p is given by

$$\Omega_p = \iint_{4\pi} F(\theta,\phi)\, d\Omega = \int_{\phi=0}^{2\pi}\left[\int_{\theta=0}^{\pi/2} \cos^2\theta \sin\theta\, d\theta\right] d\phi$$
$$= \int_{\phi=0}^{2\pi}\left[-\frac{\cos^3\theta}{3}\right]_0^{\pi/2} d\phi$$
$$= \int_0^{2\pi} \frac{1}{3}\, d\phi = \frac{2\pi}{3} \quad \text{(sr)}.$$

(c) Application of Eq. (9.23) gives

$$D = \frac{4\pi}{\Omega_p} = 4\pi\left(\frac{3}{2\pi}\right) = 6,$$

which corresponds to D (dB) $= 10\log 6 = 7.78$ dB.

(d) The half-power beamwidth β is obtained by setting $F(\theta) = 0.5$. That is,

$$F(\theta) = \cos^2\theta = 0.5,$$

which gives the half-power angles $\theta_1 = -45°$ and $\theta_2 = 45°$. Hence,

$$\beta = \theta_2 - \theta_1 = 90°.$$

Example 9-2: Directivity of a Hertzian Dipole

Calculate the directivity of a Hertzian dipole.

Solution: Application of Eq. (9.23) with $F(\theta) = \sin^2\theta$ [from Eq. (9.15)] gives

$$D = \frac{4\pi}{\displaystyle\iint_{4\pi} F(\theta,\phi)\sin\theta\, d\theta\, d\phi} = \frac{4\pi}{\displaystyle\int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} \sin^3\theta\, d\theta\, d\phi} = \frac{4\pi}{8\pi/3} = 1.5$$

or, equivalently, 1.76 dB.

9-2.4 Antenna Gain

Of the total power P_t (transmitter power) supplied to the antenna, a part, P_{rad}, is radiated out into space, and the remainder, P_{loss}, is dissipated as heat in the antenna structure. The ***radiation efficiency*** ξ is defined as the ratio of P_{rad} to P_t:

$$\xi = \frac{P_{rad}}{P_t}. \quad \text{(dimensionless)} \tag{9.27}$$

The ***gain*** of an antenna is defined as

$$G = \frac{4\pi R^2 S_{max}}{P_t}, \tag{9.28}$$

which is similar in form to the expression given by Eq. (9.24) for the directivity D except that it is referenced to the input

虽然是近似的，但这种关系提供了一种有用的方法，可以通过测量两个正交平面（其交点为主瓣轴）的波束宽度来估计天线方向性。

例 9-1： 天线辐射特性

确定仅辐射到上半球的天线的 (a) 最大辐射方向、(b) 方向图立体角、(c) 方向性和 (d) y-z 平面中的半功率波束宽度，归一化辐射强度由 $F(\theta,\phi) = \cos^2\theta$ 给出。

解：天线仅通过上半球辐射的说法等价于

$$F(\theta,\phi) = F(\theta) = \begin{cases} \cos^2\theta & \text{for } 0 \le \theta \le \pi/2 \\ & \text{and } 0 \le \phi \le 2\pi, \\ 0 & \text{elsewhere.} \end{cases}$$

(a) 函数 $F(\theta) = \cos^2\theta$ 与 ϕ 无关，并且在 $\theta = 0°$ 时最大。$F(\theta)$ 的极坐标图如图 9-13 所示。

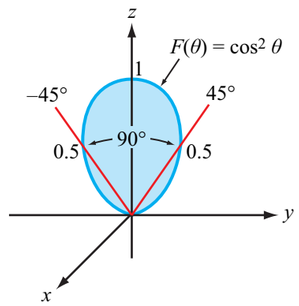


图 9-13 $F(\theta) = \cos^2\theta$ 的极坐标图。

(b) 从方程式（9.21），图案立体角Ω_p由下式给出

$$\Omega_p = \iint_{4\pi} F(\theta,\phi)\, d\Omega = \int_{\phi=0}^{2\pi}\left[\int_{\theta=0}^{\pi/2} \cos^2\theta \sin\theta\, d\theta\right] d\phi$$
$$= \int_{\phi=0}^{2\pi}\left[-\frac{\cos^3\theta}{3}\right]_0^{\pi/2} d\phi$$
$$= \int_0^{2\pi} \frac{1}{3}\, d\phi = \frac{2\pi}{3} \quad \text{(sr)}.$$

(c) 等式的应用(9.23) 给出

$$D = \frac{4\pi}{\Omega_p} = 4\pi\left(\frac{3}{2\pi}\right) = 6,$$

对应于 D (dB) = 10 log6 = 7.78 dB。

(d)通过将F(θ)=设置为0.5来获得半功率波束宽度β。那是

$$F(\theta) = \cos^2\theta = 0.5,$$

给出半功率角 $\theta_1 = -45°$ 和 $\theta_2 = 45°$。因此，

$$\beta = \theta_2 - \theta_1 = 90°.$$

例 9-2： 赫兹偶极子的方向性

计算赫兹偶极子的方向性。

解决方案：应用方程。 (9.23) 与 $F(\theta) = \sin^2\theta$ [来自方程。 (9.15)]给出

$$D = \frac{4\pi}{\displaystyle\iint_{4\pi} F(\theta,\phi)\sin\theta\, d\theta\, d\phi} = \frac{4\pi}{\displaystyle\int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} \sin^3\theta\, d\theta\, d\phi} = \frac{4\pi}{8\pi/3} = 1.5$$

或等效为 1.76 dB。

9-2.4 天线增益

在提供给天线的总功率 P_t（发射机功率)中，一部分 P_{rad} 辐射到空间中，其余部分 P_{loss} 在天线结构中作为热量耗散。辐射效率 ξ 定义为 P_{rad} 与 P_t 之比：

$$\xi = \frac{P_{rad}}{P_t}. \quad \text{(dimensionless)} \tag{9.27}$$

天线的增益定义为

$$G = \frac{4\pi R^2 S_{max}}{P_t}, \tag{9.28}$$

其形式与等式给出的表达式类似。 (9.24) 对于方向性 D，除了它参考输入